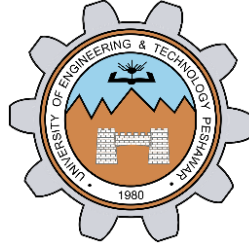


LAB # 04
INTERFACING PUSH BUTTONS TO 8051 DEVELOPMENT KIT
USING INTERRUPTS



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CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have
neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Objectives of The Lab:

- Understand the concept of interrupts.

Interrupts:

When a device needs service, it interrupts the microcontroller, which stops its current operation to handle the request. This interruption is managed by the Interrupt Service Routine (ISR), a program that addresses the device's needs. Once the ISR completes its task, the microcontroller resumes its previous operation.

Interrupts in the 8051:

- Reset
- Two interrupts for timers
- Two external hardware interrupts
- Serial communication interrupt belong to both receive and transmit

Interrupt	Source	Vector Address
System Reset	RST	0000H
External 0	IE0	0003H
Timer 0	TF0	000BH
External 1	IE1	0013H
Timer 1	TF1	001BH
Serial Port	RI or TI	0023H

IE register is used for enabling and disabling the interrupt.

This is a bit addressable register in which EA value must be set to one for enabling interrupts.

	EA	--	--	ES	ET1	EX1	ET0	EX0
EA	IE.7	Disables all interrupts, If EA=0, no interrupt will be acknowledged. If EA=1, interrupt source is individually enable or disabled by setting or clearing its enable bit.						
--	IE.6	Not implemented, reserved for future use*.						
--	IE.5	Not implemented, reserved for future use*.						
ES	IE.4	Enable or disable the Serial port interrupt.						
ET1	IE.3	Enable or disable the Timer 1 overflow interrupt.						
EX1	IE.2	Enable or disable External interrupt 1.						
ET0	IE.1	Enable or disable the Timer 0 overflow interrupt.						
EX0	IE.0	Enable or disable External interrupt 0.						

Lab Tasks:

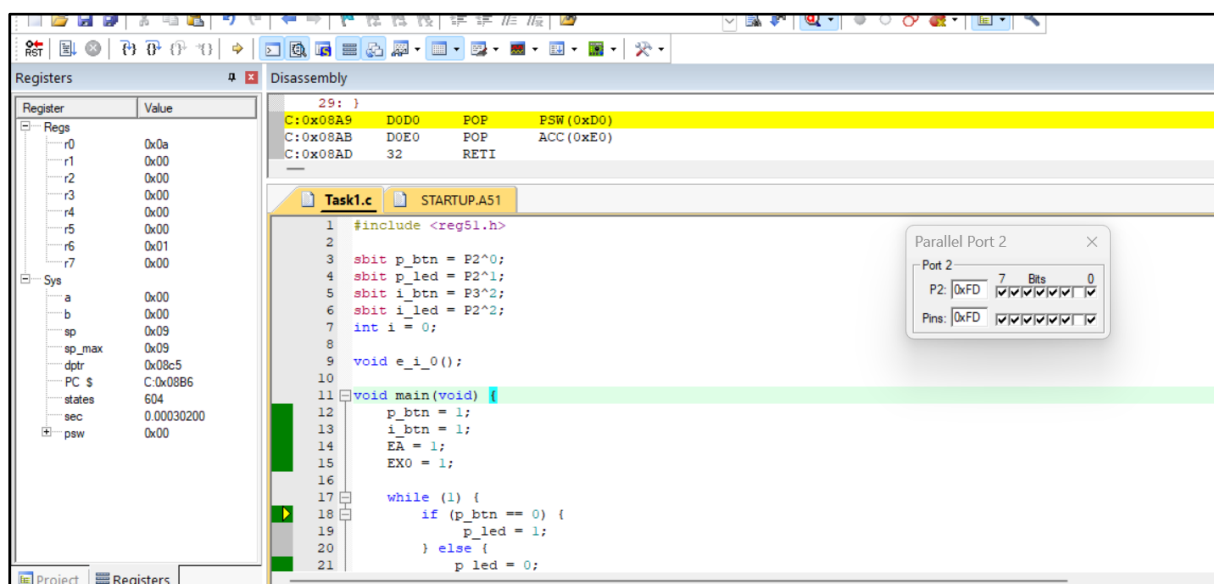
Task 1:

Task in the slide/lecture.

C Code:

```
Task1.c  STARTUP.A51
1  #include <reg51.h>
2
3  sbit p_btn = P2^0;
4  sbit p_led = P2^1;
5  sbit i_btn = P3^2;
6  sbit i_led = P2^2;
7  int i = 0;
8
9  void e_i_0();
10
11 void main(void) {
12     p_btn = 1;
13     i_btn = 1;
14     EA = 1;
15     EX0 = 1;
16
17     while (1) {
18         if (p_btn == 0) {
19             p_led = 1;
20         } else {
21             p_led = 0;
22         }
23     }
24 }
25
26 void e_i_0() interrupt 0 {
27     for (i = 0; i < 100; i++);
28     i_led ^= 1;
29 }
30
```

Keil Simulation:



The screenshot shows the Keil uVision IDE in simulation mode. The main window displays the C code from Task1.c. The left pane shows the 'Registers' window with the following values:

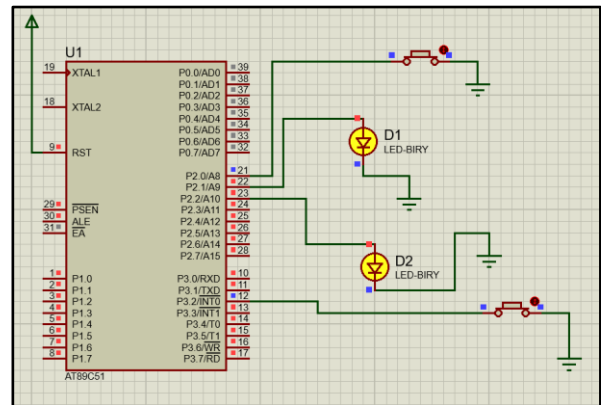
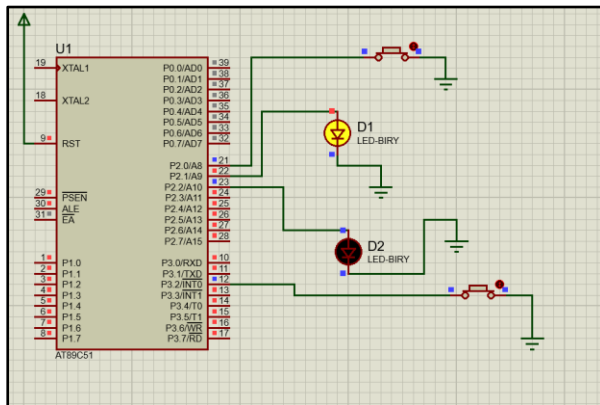
Register	Value
r0	0x0a
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x01
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x09
sp_max	0x09
dptr	0x08c5
PC	0x08B6
states	604
sec	0.00030200
psw	0x00

The right pane shows the 'Disassembly' window with the following instructions:

Address	Disassembly
C:0x08A9	D0D0 POP FSW (0xD0)
C:0x08AB	D0E0 POP ACC (0xE0)
C:0x08AD	32 RETI

A 'Parallel Port 2' dialog box is open, showing the port configuration. The 'Port 2' tab is selected, and the 'P2' value is 0xFD. The 'Pins' tab is also visible, showing the pin configuration.

Proteus Simulation:



Task 2:

Attach 7seg-BCD with PORT1 and whenever interrupt occur at PORT3 ($P3^2$) display in the seven segment. On pressing the button, it displays 1 again pressing the button it should display 2. **Hint:** Increment PORT1 when an interrupt occurs.

C Code:

```

1  #include <reg51.h>
2  sbit i_btn = P3^2;
3
4  void e_i_0();
5
6  void main(void) {
7      i_btn = 1;
8      EA = 1;
9      EX0 = 1;
10     IT0 = 1;
11 }
12
13 void e_i_0() interrupt 0 {
14     P1++;
15 }
16

```

Keil Simulation:

The Keil uVision simulation interface displays the following components:

- Registers:** A table showing the state of registers. The 'sp' register is highlighted with a value of 0x07.
- Disassembly:** A list of assembly instructions. The instruction `C:0x080E D2AF SETB EA(0xA8.7)` is highlighted.
- Source Code:** The C code for `7seg_interrupt.c` is shown. The line `EA = 1;` is highlighted.
- Parallel Port 1:** A configuration window for Port 1, showing P1: 0x08 and Pins: 0x08.
- Parallel Port 3:** A configuration window for Port 3, showing P3: 0x04 and Pins: 0x04.
- Interrupt System:** A table showing interrupt sources and their configurations. The 'EA' (Enable All) checkbox is checked.

Int Source	Vector	Mode	Req	Ena	Pri
P3.2/int0	0003H	1	0	1	0
Timer 0	000BH	0	0	0	0
P3.3/int1	0013H	0	1	0	0
Timer 1	001BH	0	0	0	0
Serial Rcv.	0023H	0	0	0	0
Serial Xmt.	0023H	0	0	0	0

Selected Interrupt	
☒ EA	☒ IT0 ☐ IEO ☒ EX0 Pri.: 0x00

Proteus Simulation:

